CS 215: Task 03: Due: 22-Aug Time: as mentioned in Piazza

- Please write (only if true) the honor code. If you used any source (person or thing) explicitly state it. This is an individual assignment.
- Your answers should be in your handwriting.
- Problems are not necessarily in order of difficulty.

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0. Please write the honor code in your handwriting, with the understanding that it is true.

1. A person tried by a 3-judge panel is declared guilty if at least 2 judges cast votes of guilty. Suppose that when the defendant is in fact guilty, each judge will independently vote guilty with probability 0.7, whereas when the defendant is in fact innocent, this probability drops to 0.2. If 70 percent of defendants are guilty, compute (see hints ahead) the conditional probability that judge number 3 votes guilty given that

   Hint: This set of problem has three elements.
   (a) Joint probabilties, the chain rule: $P(A, B, C) = P(A|B, C)P(B|C)P(C)$, and Bayes' rule
   (b) The second line with the word "independent" is the heart of the simplification of the joint probabilities
   (c) If you get stuck trying to solve this problem, consider changing the problem to a two judge scenario, with the artificial rule that a defendent is guilty if either of the two judges declares the defendant to be guilty.

   (a) judges 1 and 2 vote guilty, Let G denote that the defendant is guilty and I denote that the defendant is innocent. We want to find

   $$P(J_3 = G \mid J_1 = G, J_2 = G).$$

   Since each judge votes independently given the defendant's status,

   $$P(J_1 = G, J_2 = G \mid G) = P(J_1 = G \mid G) \cdot P(J_2 = G \mid G) = (0.7)^2 = 0.49,$$

   and

   $$P(J_1 = G, J_2 = G \mid I) = P(J_1 = G \mid I) \cdot P(J_2 = G \mid I) = (0.2)^2 = 0.04.$$

   By Bayes' theorem,

   $$P(G \mid J_1 = G, J_2 = G) = \frac{P(J_1 = G, J_2 = G \mid G)P(G)}{P(J_1 = G, J_2 = G \mid G)P(G) + P(J_1 = G, J_2 = G \mid I)P(I)}$$

   $$= \frac{(0.49)(0.7)}{(0.49)(0.7) + (0.04)(0.3)} = \frac{0.343}{0.355} = \frac{343}{355}.$$

   Therefore,

   $$P(I \mid J_1 = G, J_2 = G) = 1 - \frac{343}{355} = \frac{12}{355}.$$

Now, using the conditional independence of judge 3's vote,

$$\begin{aligned} P(J_3 = G \mid J_1 = G, J_2 = G) &= P(J_3 = G \mid G)P(G \mid J_1 = G, J_2 = G) \\ &\quad + P(J_3 = G \mid I)P(I \mid J_1 = G, J_2 = G) \\ &= (0.7)\frac{343}{355} + (0.2)\frac{12}{355} \\ &= \frac{242.5}{355} \\ &\approx 0.6831. \end{aligned}$$

Hence,

$$\boxed{P(J_3 = G \mid J_1 = G, J_2 = G) \approx 0.6831}.$$

(b) judges 1 and 2 cast one guilty and one not guilty vote, We want to find

$$P(J_3 = G \mid J_1 = G, J_2 = I).$$

Given that the defendant is guilty,

$$P(J_1 = G, J_2 = I \mid G) = (0.7)(0.3) = 0.21.$$

Given that the defendant is innocent,

$$P(J_1 = G, J_2 = I \mid I) = (0.2)(0.8) = 0.16.$$

By Bayes' theorem,

$$P(G \mid J_1 = G, J_2 = I) = \frac{(0.21)(0.7)}{(0.21)(0.7) + (0.16)(0.3)} = \frac{0.147}{0.195} = \frac{49}{65}.$$

Thus,

$$P(I \mid J_1 = G, J_2 = I) = 1 - \frac{49}{65} = \frac{16}{65}.$$

Therefore,

$$\begin{aligned} P(J_3 = G \mid J_1 = G, J_2 = I) &= P(J_3 = G \mid G)P(G \mid J_1 = G, J_2 = I) \\ &\quad + P(J_3 = G \mid I)P(I \mid J_1 = G, J_2 = I) \\ &= (0.7)\frac{49}{65} + (0.2)\frac{16}{65} \\ &= \boxed{\frac{37.5}{65} \approx 0.5769}. \end{aligned}$$

(c) judges 1 and 2 both cast not guilty votes. We want to find

$$P(J_3 = G \mid J_1 = I, J_2 = I).$$

Given that the defendant is guilty,

$$P(J_1 = I, J_2 = I \mid G) = (0.3)^2 = 0.09.$$

Given that the defendant is innocent,

$$P(J_1 = I, J_2 = I \mid I) = (0.8)^2 = 0.64.$$

By Bayes' theorem,

$$P(G \mid J_1 = I, J_2 = I) = \frac{(0.09)(0.7)}{(0.09)(0.7) + (0.64)(0.3)} = \frac{0.063}{0.255} = \frac{21}{85}.$$

Thus,

$$P(I \mid J_1 = I, J_2 = I) = 1 - \frac{21}{85} = \frac{64}{85}.$$

Therefore,

$$\begin{aligned} P(J_3 = G \mid J_1 = I, J_2 = I) &= P(J_3 = G \mid G)P(G \mid J_1 = I, J_2 = I) \\ &\quad + P(J_3 = G \mid I)P(I \mid J_1 = I, J_2 = I) \\ &= (0.7)\frac{21}{85} + (0.2)\frac{64}{85} \\ &= \boxed{\frac{27.5}{85} \approx 0.3235}. \end{aligned}$$

Observe that as the evidence shifts based on other judges, Judge 3's confidence changes.

2. Let X and Y be random variables with finite variances

$$\mathrm{Var}(X) = \sigma_X^2 \qquad \text{and} \qquad \mathrm{Var}(Y) = \sigma_Y^2.$$

(a) As something unrelated to X and Y – but will be used later on in (d) – prove that if $\mathrm{Var}(Z) = 0$, then

$$P\{Z = \mathbb{E}[Z]\} = 1.$$

By definition of variance,

$$\mathrm{Var}(Z) = \mathbb{E}\left[(Z - \mathbb{E}[Z])^2\right].$$

Since $\mathrm{Var}(Z) = 0$, we have

$$\mathbb{E}\left[(Z - \mathbb{E}[Z])^2\right] = 0.$$

Now, $(Z - \mathbb{E}[Z])^2 \geq 0$. A nonnegative random variable can have expectation 0 only if it is equal to 0 almost surely.

Aside. The phrase "almost surely": Let $W = (Z - \mathbb{E}[Z])^2$ be non-negative. Piazza post 53 https://piazza.com/class/ms1ll7021ih5yc/post/53_f3 is pointing to the fact that if the expectation – written as an integral – is zero does not mean that the integrand is zero.
But let's say the integrand is non-zero. If W happens to be non-zero with probability > 0 on some weird situation, then the expectation will be non-zero. So the probability that this happens is 0, and thus W = 0, almost surely.

Hence,

$$(Z - \mathbb{E}[Z])^2 = 0$$

Therefore,

$$Z = \mathbb{E}[Z] (\text{almost surely})$$

which means

$$\boxed{P\{Z = \mathbb{E}[Z]\} = 1}.$$

(b) Using the fact that variance is always nonnegative, show that

$$0 \leq \text{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right)$$

implies that

$$-1 \leq \text{Corr}(X, Y).$$

Using the properties of variance,

$$\begin{aligned} 0 &\leq \text{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right) \\ &= \frac{\text{Var}(X)}{\sigma_X^2} + \frac{\text{Var}(Y)}{\sigma_Y^2} + \frac{2\,\text{Cov}(X, Y)}{\sigma_X \sigma_Y}. \end{aligned}$$

Since $\sigma_X^2 = \text{Var}(X)$ and $\sigma_Y^2 = \text{Var}(Y)$,

$$0 \leq 1 + 1 + 2\frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Using

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y},$$

we obtain

$$0 \leq 2 + 2\,\text{Corr}(X, Y).$$

and hence

$$\boxed{-1 \leq \text{Corr}(X, Y)}.$$

(c) Similarly, use

$$0 \leq \text{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right)$$

to prove that

$$-1 \leq \text{Corr}(X, Y) \leq 1.$$

Using the properties of variance,

$$\begin{aligned} 0 &\leq \text{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right) \\ &= \frac{\text{Var}(X)}{\sigma_X^2} + \frac{\text{Var}(Y)}{\sigma_Y^2} - \frac{2\,\text{Cov}(X, Y)}{\sigma_X \sigma_Y}. \end{aligned}$$

Since $\sigma_X^2 = \text{Var}(X)$ and $\sigma_Y^2 = \text{Var}(Y)$,

$$0 \leq 1 + 1 - 2\frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Using

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y},$$

we obtain

$$0 \leq 2 - 2\,\mathrm{Corr}(X, Y).$$

and hence

$$\boxed{\mathrm{Corr}(X, Y) \leq 1}.$$

(d) Using the result from part (a), prove that if

$$\mathrm{Corr}(X, Y) = 1$$

or

$$\mathrm{Corr}(X, Y) = -1,$$

then X and Y must satisfy a linear relationship of the form

$$Y = a + bX,$$

where $b > 0$ when $\mathrm{Corr}(X, Y) = 1$ and $b < 0$ when $\mathrm{Corr}(X, Y) = -1$.
If $\mathrm{Corr}(X, Y) = 1$, then equality holds in the variance inequality proved earlier, and hence

$$\mathrm{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right) = 0.$$

By part (a), if a random variable has variance 0, then it is equal to its expectation almost surely. Hence,

$$\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y} = \mathbb{E}\left[\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right]$$

Therefore,

$$Y = \frac{\sigma_Y}{\sigma_X} X + \sigma_Y \left(\frac{\mathbb{E}[Y]}{\sigma_Y} - \frac{\mathbb{E}[X]}{\sigma_X}\right).$$

Thus,

$$Y = a + bX, \qquad b = \frac{\sigma_Y}{\sigma_X} > 0$$

since Corr(X,y) is well defined. Similarly, if $\mathrm{Corr}(X, Y) = -1$, then

$$\mathrm{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right) = 0.$$

Hence,

$$\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y} = \mathbb{E}\left[\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right]$$

$$Y = -\frac{\sigma_Y}{\sigma_X} X + \sigma_Y \left(\frac{\mathbb{E}[X]}{\sigma_X} + \frac{\mathbb{E}[Y]}{\sigma_Y}\right).$$

Thus,

$$Y = a + bX, \qquad b = -\frac{\sigma_Y}{\sigma_X} < 0.$$

Therefore, in either case, X and Y satisfy a linear relationship

$$\boxed{Y = a + bX},$$

with $b > 0$ when $\mathrm{Corr}(X, Y) = 1$ and $b < 0$ when $\mathrm{Corr}(X, Y) = -1$.

3. Consider n independent trials, each of which results in one of the three outcomes 1, 2, or 3, with corresponding probabilities p_1, p_2, and p_3, where

$$p_1 + p_2 + p_3 = 1.$$

Let N_i denote the number of trials that result in outcome i.

Find

$$\mathbb{E}[N_i]$$

For each $j = 1, 2, \ldots, n$, define the indicator random variable

$$I_{ij} = \begin{cases} 1, & \text{if the } j\text{-th trial results in outcome } i, \\ 0, & \text{otherwise.} \end{cases}$$

Since N_i is the total number of trials that result in outcome i,

$$N_i = \sum_{j=1}^{n} I_{ij}.$$

For each trial j,

$$\mathbb{P}(I_{ij} = 1) = p_i, \qquad \mathbb{P}(I_{ij} = 0) = 1 - p_i.$$

Therefore,

$$\begin{aligned} \mathbb{E}[I_{ij}] &= 1 \cdot \mathbb{P}(I_{ij} = 1) + 0 \cdot \mathbb{P}(I_{ij} = 0) \\ &= 1 \cdot p_i + 0 \cdot (1 - p_i) \\ &= p_i. \end{aligned}$$

Using linearity of expectation,

$$\begin{aligned} \mathbb{E}[N_i] &= \mathbb{E}\left[\sum_{j=1}^{n} I_{ij}\right] \\ &= \sum_{j=1}^{n} \mathbb{E}[I_{ij}] \\ &= \sum_{j=1}^{n} p_i \\ &= \boxed{np_i}. \end{aligned}$$

4. Let (X, Y, Z) have joint probability density

$$f_{X,Y,Z}(x, y, z) = \begin{cases} c(x + z), & 0 < x < y < z < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Determine the constant c.

Since $f_{X,Y,Z}$ is a probability density function, its integral over the entire support must be equal to 1. Hence,

$$\int_0^1 \int_0^z \int_{y=x}^{z} c(x + z)\, dy\, dx\, dz = 1.$$

Integrating first with respect to y,

$$c\int_0^1\int_0^z (x+z)(z-x)\,dx\,dz = 1.$$

Since

$$(x+z)(z-x) = z^2 - x^2,$$

we get

$$c\int_0^1\int_0^z (z^2-x^2)\,dx\,dz = c\int_0^1 \frac{2z^3}{3}\,dz = \frac{c}{6}.$$

Therefore,

$$\boxed{c = 6}.$$

(b) Compute

$$P(X+Y<Z).$$

Hint: Refer to example 4.3c from the textbook

We need the region satisfying both

$$0 < x < y < z < 1$$

and

$$x + y < z.$$

For fixed z, since $y > x$ and $y < z - x$, we must have

$$x < z - x,$$

or equivalently,

$$0 < x < \frac{z}{2}.$$

Thus,

$$P(X+Y<Z) = 6\int_0^1\int_0^{z/2}\int_x^{z-x} (x+z)\,dy\,dx\,dz.$$

Integrating with respect to y,

$$= 6\int_0^1\int_0^{z/2} (x+z)(z-2x)\,dx\,dz.$$

Evaluating the remaining integrals gives

$$P(X+Y<Z) = 6\int_0^1 \frac{7z^3}{24}\,dz = \frac{7}{16}.$$

Hence,

$$\boxed{P(X+Y<Z) = \frac{7}{16}}.$$

(c) Compute

$$E[XZ].$$

Using the definition of expectation for jointly continuous random variables,

$$E[XZ] = \iiint xz\, f_{X,Y,Z}(x,y,z)\, dydxdz.$$

Therefore,

$$E[XZ] = 6\int_0^1 \int_0^z \int_x^z xz(x+z)\, dy\, dx\, dz.$$

Integrating first with respect to y,

$$E[XZ] = 6\int_0^1 \int_0^z xz(x+z)(z-x)\, dx\, dz.$$

Since

$$(x+z)(z-x) = z^2 - x^2,$$

this becomes

$$6\int_0^1 \int_0^z xz(z^2-x^2)\, dx\, dz.$$

Evaluating,

$$E[XZ] = 6\int_0^1 \frac{z^5}{4}\, dz = \frac{1}{4}.$$

Hence,

$$\boxed{E[XZ] = \frac{1}{4}}.$$

5. Let $X \geq 0$ be a random variable satisfying

$$E[X] = 4, \qquad E[X^2] = 24, \qquad E[X^3] = 192.$$

Use Markov's inequality, applied appropriately to X, X^2, and X^3, to obtain a tight bound from the given information on

$$P(X \geq 10).$$

Since $X \geq 0$, Markov's inequality gives

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

Applying it directly to X,

$$P(X \geq 10) \leq \frac{E[X]}{10} = \frac{4}{10} = 0.4.$$

We can also apply Markov's inequality to the nonnegative random variable X^2. Since

$$X \geq 10 \quad \Longleftrightarrow \quad X^2 \geq 100,$$

we get

$$P(X \geq 10) = P(X^2 \geq 100) \leq \frac{E[X^2]}{100} = \frac{24}{100} = 0.24.$$

Similarly, applying Markov's inequality to X^3,

$$P(X \geq 10) = P(X^3 \geq 1000) \leq \frac{E[X^3]}{1000} = \frac{192}{1000} = 0.192.$$

Therefore, the tightest of the three bounds is

$$\boxed{P(X \geq 10) \leq 0.192}.$$

A linear combination, or more generally, a non-linear combination, of the given expectations is possible to get an even tighter bound. This is a much more open ended problem as also noted in the Piazza post.

6. Let $X > 0$ be a random variable satisfying

$$E[X] = \mu.$$

For some constant $a > 0$, define

$$g(x) = x + \frac{a}{x}.$$

(a) Use Jensen's inequality to obtain a bound, in terms of μ and a, for

$$E\left[X + \frac{a}{X}\right].$$

For $x > 0$,

$$g(x) = x + \frac{a}{x}.$$

Since

$$g''(x) = \frac{2a}{x^3} > 0,$$

the function g is convex on $(0, \infty)$.
Therefore, by Jensen's inequality,

$$E[g(X)] \geq g(E[X]).$$

Using $E[X] = \mu$,

$$E\left[X + \frac{a}{X}\right] \geq \mu + \frac{a}{\mu}.$$

Hence,

$$\boxed{E\left[X + \frac{a}{X}\right] \geq \mu + \frac{a}{\mu}}.$$

(b) If μ may also be chosen freely, determine the value of μ to minimize the Jensen bound.

From part (a), the Jensen bound is

$$\mu + \frac{a}{\mu}, \qquad \mu > 0.$$

To minimize this expression, differentiate with respect to μ:

$$\frac{d}{d\mu}\left(\mu + \frac{a}{\mu}\right) = 1 - \frac{a}{\mu^2}.$$

Setting the derivative equal to zero gives

$$1 - \frac{a}{\mu^2} = 0,$$

so

$$\mu^2 = a.$$

Since $\mu > 0$,

$$\boxed{\mu = \sqrt{a}}.$$

At this value, the minimum Jensen bound is

$$\sqrt{a} + \frac{a}{\sqrt{a}} = 2\sqrt{a}.$$

Thus,

$$\boxed{\min_{\mu>0} \left(\mu + \frac{a}{\mu}\right) = 2\sqrt{a}}.$$

7. Let $X_1, \ldots, X_{25}$ be identically distributed random variables with

$$E[X_i] = 100, \qquad \mathrm{Var}(X_i) = 400.$$

Suppose that, for $i \neq j$,

$$\mathrm{Cov}(X_i, X_j) = 40.$$

Let

$$\overline{X} = \frac{1}{25} \sum_{i=1}^{25} X_i.$$

(a) Use Chebyshev's inequality to obtain a lower bound for

$$P(92 < \overline{X} < 108).$$

First,

$$E[\overline{X}] = \frac{1}{25} \sum_{i=1}^{25} E[X_i] = 100.$$

To compute the variance, we use

$$\mathrm{Var}\left(\sum_{i=1}^{25} X_i\right) = \sum_{i=1}^{25} \mathrm{Var}(X_i) + \sum_{i \neq j} \mathrm{Cov}(X_i, X_j).$$

Hence,

$$\begin{aligned} \mathrm{Var}(\overline{X}) &= \frac{1}{25^2} \left(25(400) + 25(24)(40)\right) \\ &= \frac{34000}{625} = 54.4. \end{aligned}$$

Now,

$$P(92 < \overline{X} < 108) = P(|\overline{X} - 100| < 8).$$

By Chebyshev's inequality,

$$P(|\overline{X} - 100| \geq 8) \leq \frac{\text{Var}(\overline{X})}{8^2} = \frac{54.4}{64} = 0.85.$$

Therefore,

$$\boxed{P(92 < \overline{X} < 108) \geq 0.15}.$$

(b) Compare this bound with the one obtained if the X_i's were independent. Expand on your observations on the the two differing bounds.

If the X_i's were independent, then all the covariance terms would be zero. Thus,

$$\text{Var}(\overline{X}) = \frac{1}{25^2}\left(25(400)\right) = 16.$$

Applying Chebyshev's inequality again,

$$P(|\overline{X} - 100| \geq 8) \leq \frac{16}{64} = \frac{1}{4}.$$

Hence,

$$\boxed{P(92 < \overline{X} < 108) \geq \frac{3}{4}}.$$

Thus, the Chebyshev lower bound increases from

$$0.15 \qquad \text{to} \qquad 0.75$$

when the variables are independent.

The reason the lower bound tells us more about the probabilistic event is obviously because the variance has dropped dramatically from ≈54 to 16. When the observations are independent, averaging reduces variability. If this was a real experiment with 25 different students, we would not want the students to "copy" the results from one another (larger covariance), but be more "independent" (covariance = 0).